Robot Dynamics Lab
Exercise 5: Joint Control Test Bench

Your Name Here

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Part A: Analytical Controller Design

This section details the analytical design of the cascade controller for the robot joint module, following the tasks in the exercise sheet.

1.1 Control Condition

Task 1: State the control law for U_d.

Solution: The system equation (1) is $0 = U_d + p\omega_m L_q I_q$. To ensure $I_d = 0$ in the d-q reference frame, the control law for U_d must be:

$$U_d = -p\omega_m L_q I_q$$

1.2 Back-EMF Compensation

Task 2: Determine the compensation voltage $U_{q,komp}$ to counteract the back-electromotive force (back-EMF).

Solution: The full equation for the q-axis voltage is $L_q \frac{dI_q}{dt} = U_q - RI_q - p\omega_m \psi_{PM}$. We want to simplify this to $\hat{U}_q = L_q \frac{dI_q}{dt} + RI_q$ using the relation $U_q = \hat{U}_q + U_{q,komp}$. By comparing the two equations, the back-EMF term must be cancelled by the compensation voltage. Therefore:

$$U_{q,komp} = p\omega_m \psi_{PM}$$

1.3 Current Controller Design

Task 3.1-3.3: Plant Transfer Function. The plant for the current control loop consists of the power amplifier ($H_{LV}(s)$) and the electrical part of the motor ($H_{el}(s)$) in series.

1. **Motor electrical part ($H_{el}(s)$):** From the simplified equation $\hat{U}_q = L_q \frac{dI_q}{dt} + RI_q$, the Laplace transform gives $\hat{U}_q(s) = (sL_q + R)I_q(s)$. The transfer function is a PT1 element:

$$H_{el}(s) = \frac{I_q(s)}{\hat{U}_q(s)} = \frac{1}{sL_q + R} = \frac{1/R}{1 + s(L_q/R)} = \frac{1/R}{1 + sT_{el}}$$

where $T_{el} = L_q/R$.

2. **Power amplifier ($H_{LV}(s)$):** This is approximated as a PT1 element with gain 1 and time constant T_{LV}:
$$H_{LV}(s) = \frac{1}{1 + sT_{LV}}$$

3. **Total plant ($G_I(s)$):** The series connection gives a PT2 element:
$$G_I(s) = H_{LV}(s) \cdot H_{el}(s) = \frac{1/R}{(1 + sT_{LV})(1 + sT_{el})}$$

Task 3.4: PI Controller Design. We design a PI controller based on the **Magnitude Optimum (BO)** criterion. For a PT2 plant, we use row 3 of the optimization table. Given $T_{el} > T_{LV}$, we set $T_1 = T_{el}$ and $T_\sigma = T_{LV}$. The plant gain is $V_S = 1/R$. The controller parameters are:
$$V_{IR} = \frac{T_1}{2T_\sigma V_S} = \frac{T_{el}}{2 \cdot T_{LV} \cdot (1/R)} = \frac{R \cdot T_{el}}{2T_{LV}}$$
$$T_{In} = T_1 = T_{el}$$

1.4 Speed Controller Design

Prerequisite: The closed-loop current controller $K_I(s)$ is approximated as a PT1 element: $K_I(s) \approx \frac{1}{1+s\cdot 2T_{LV}}$.

Task 4.1-4.2: Plant Transfer Function. The plant for the speed loop consists of the closed-loop current controller ($K_I(s)$) and the mechanical part of the motor ($H_\omega(s)$).

1. **Motor mechanical part ($H_\omega(s)$):** From equations $M_m = \frac{3}{2}p\psi_{PM}I_q$ and $J\frac{d\omega_m}{dt} = M_m$ (ignoring disturbance M_{Lm}), we get $Js\omega_m(s) = \frac{3}{2}p\psi_{PM}I_q(s)$. The transfer function is an I-element:
$$H_\omega(s) = \frac{\omega_m(s)}{I_q(s)} = \frac{\frac{3}{2}p\psi_{PM}}{sJ}$$

2. **Total plant ($G_\omega(s)$):** The series connection gives an IT1 element:
$$G_\omega(s) = K_I(s) \cdot H_\omega(s) = \frac{1}{1 + s \cdot 2T_{LV}} \cdot \frac{\frac{3}{2}p\psi_{PM}}{sJ}$$

Task 4.3: PI Controller Design. We design a PI controller based on the **Symmetrical Optimum (SO)** criterion. For an IT1 plant, we use row 9 of the optimization table. Comparing $G_\omega(s)$ with the standard form $\frac{V_S}{sT_1(1+sT_\sigma)}$, we identify $T_\sigma = 2T_{LV}$ and $\frac{V_S}{T_1} = \frac{\frac{3}{2}p\psi_{PM}}{J}$. The parameters are:
$$V_{\omega R} = \frac{T_1}{2T_\sigma V_S} = \frac{1}{2T_\sigma}\left(\frac{T_1}{V_S}\right) = \frac{1}{2(2T_{LV})}\left(\frac{J}{\frac{3}{2}p\psi_{PM}}\right) = \frac{J}{6T_{LV}p\psi_{PM}}$$
$$T_{\omega n} = 4T_\sigma = 4 \cdot (2T_{LV}) = 8T_{LV}$$

1.5 Position Controller Design

Prerequisite: The closed-loop speed controller $K_\omega(s)$ is approximated as a PT1 element: $K_\omega(s) \approx \frac{1}{1+s\cdot 8T_{LV}}$.

Task 5.1-5.2: Plant Transfer Function. The plant for the position loop consists of the closed-loop speed controller ($K_\omega(s)$) and the integration from speed to position ($H_\varphi(s)$).

1. **Position integration ($H_\varphi(s)$):** From $\omega_m = \frac{d\varphi_m}{dt}$, the transfer function is a pure I-element:

$$H_\varphi(s) = \frac{\varphi_m(s)}{\omega_m(s)} = \frac{1}{s}$$

2. **Total plant ($G_\varphi(s)$):** The series connection gives an IT1 element:

$$G_\varphi(s) = K_\omega(s) \cdot H_\varphi(s) = \frac{1}{1 + s \cdot 8T_{LV}} \cdot \frac{1}{s}$$

Task 5.2: P and PI Controller Design.

- **Variant 1 (P Controller, BO):** For an IT1 plant, we use row 8 of the optimization table. We identify $T_\sigma = 8T_{LV}$ and $T_1/V_S = 1$.

$$V_{\varphi R}^{P} = \frac{T_1}{2T_\sigma V_S} = \frac{1}{2T_\sigma}\left(\frac{T_1}{V_S}\right) = \frac{1}{2(8T_{LV})}(1) = \frac{1}{16T_{LV}}$$

- **Variant 2 (PI Controller, SO):** For an IT1 plant, we use row 9.

$$V_{\varphi R}^{PI} = \frac{T_1}{2T_\sigma V_S} = \frac{1}{16T_{LV}}$$

$$T_{\varphi n}^{PI} = 4T_\sigma = 4 \cdot (8T_{LV}) = 32T_{LV}$$

Part B: Simulation Analysis

This section provides concise answers to the analysis questions based on the simulation results.

Task 7.2: P vs. PI Position Controller

Question: What changed in the control behavior and why?

Answer:

- **Observation:** When using the P-controller (Variant 1) under load, a constant steady-state error of approx. 0.024 degrees is observed. With the PI-controller (Variant 2), this steady-state error is eliminated and converges to zero.
- **Justification:** This is due to the integral (I) component. A P-controller's output is proportional to the error, so a constant error is required to counteract a constant disturbance (like the gravity load). The I-component of the PI-controller integrates the error over time, continuously increasing the control output until the error is completely compensated.

Task 7.3: Effect of Measurement Noise

Question: How do you assess the control behavior? How could you counteract this?

Answer:

- **Assessment:** The control behavior is significantly degraded by measurement noise. It introduces high-frequency oscillations into the system, visible in both the joint angle and the control signals. On real hardware, this would cause vibrations and increase mechanical wear.
- **Counter-measure:** A low-pass filter can be added to the sensor feedback path to attenuate the high-frequency noise. This creates a trade-off, as filtering can slow down the system's response time.

Task 7.4: Effect of Increased Position Sensor Noise

Question: What do you observe? What is the critical value for the parameter `r_abs`?

Answer:

- **Observation:** When the noise parameter `r_abs` is increased to `5e-5`, the system becomes completely unstable. The joint angle oscillates with a rapidly growing amplitude and completely fails to follow the setpoint.
- **Justification:** The system's stability is highly sensitive to sensor signal quality. The increased noise is amplified by the derivative actions within the control loops (especially the speed controller), causing the control signals to saturate and ultimately destabilizing the system. The critical value is the noise level at which the system's stability margin is exceeded.